MIL-F-44282(GL)
25 June 1987

MILITARY SPECIFICATION

FRUIT SOUP, ARTIFICIALLY FLAVORED (Operational Ration Component)

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers fruit soup in the form of dry powder for use by the Department of Defense as a component of the Ration, Cold Weather.

1.2 Classification. The fruit soup shall be of the following types as specified (see 6.1):

Type I - Strawberry
Type II - Raspberry

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-44276 - Packaging of Ration, Cold Weather Components in Flexible Pouches.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8915

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-22T01:10Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Application for copies should be addressed to the Office of Drinking Water WH550, Environmental Protection Agency, Washington, DC 20640.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:10Z
Check the source to verify that this is the current version before use.

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(*Application for copies should be addressed to the National Academy Press*, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.2).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Sugar, white, bakers special. Sugar shall be white, refined, granulated cane or beet sugar, or a combination thereof, and shall comply with MIL-STD-900.

3.2.2 Dextrose. The free-flowing dextrose hydrate powder shall comply with the Food Chemicals Codex.

3.2.3 Starch, gelatinized. The starch shall be a precooked, waxy-maize starch that demonstrates rapid viscosity development without cooking and attains complete viscosity development in both acid and neutral pH foods. The starch shall be a powder with a moisture content not more than 6 percent, an ash content of 0.5 percent, and a pH between 4.8 and 5.2. A mixture of 5 percent Dry Starch Base (DSB), 15 percent sucrose, and 80 percent water shall yield a viscosity of about 1000 Brabender units at 25°C after 15 minutes in a buffer of 3.5 pH. The same mixture shall yield a viscosity of about 790 Brabender units after 10 minutes at 25°C in a 6.5 pH buffer.

3.2.4 Gum arabic, powdered. Powdered gum arabic shall comply with the requirements of the Food Chemical's Codex.

3.2.5 Fumaric acid, powdered. Powdered fumaric acid shall comply with the requirements of the Food Chemicals Codex.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:10Z
Check the source to verify that this is the current version before use.

3.2.6 Ascorbic acid, fine granulated. Fine granulated ascorbic acid shall comply with the requirements of the Food Chemical Codex. Ascorbic acid content shall not be less than 90.0 percent.

3.2.7 Flavoring, strawberry, artificial. Artificial strawberry flavoring shall be equal to Caps (R) Artificial Strawberry No. 1979 manufactured by Borden or shall be equal to Artificial Strawberry #282-07-01/TP 05.00 manufactured by Firmenich.

3.2.8 Flavoring, raspberry, artificial. Artificial raspberry flavoring shall be equal to Artificial Raspberry #59.434/TP 05.50 manufactured by Firmenich or shall be equal to Caps (R) Artificial Raspberry Flavor No. 2397 manufactured by Borden.

3.2.9 Color, Red, No. 40. Red No. 40 shall comply with the requirements of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199).

3.2.10 Color, Blue No.1. Blue No. 1 color shall comply with the requirements of the Federal Food, Drug, and Cosmetic Act and the regulations promulgated thereunder (21 CFR Parts 1-199).

3.3 Preparation and processing.

3.3.1 Preparation of fruit soup base. The following ingredients shall be mixed until homogeneous:

Ingredients	Percent by Weight
Sugar	65.8
Starch	16.5
Dextrose	13.5
Gum arabic	3.0
Fumaric acid	1.0
Ascorbic acid	0.2

3.3.2 Preparation of strawberry (type I) fruit soup dry mix.

Ingredient	Percent by weight	
	Formulation A	Formulation B
Fruit soup base (see 3.3.1)	99.57	99.83
Flavoring, Caps (R) Artificial Strawberry No. 1979 (Borden) or equal	0.36	-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:10Z
Check the source to verify that this is the current version before use.

Ingredient	Percent by weight Formulation A	Formulation B
Flavoring, Artificial Strawberry #282-07-01/TP 05.00 (Firmenich) or equal	-	0.10
Color, Red No. 40 (see 6.3)	0.07	0.07

Layer the flavoring and the color between two layers of the fruit soup base and mix until homogeneous.

3.3.3 Preparation of raspberry (type II) fruit soup dry mix.

Ingredient	Percent by weight Formulation A	Formulation B
Fruit soup base (see 3.3.1)	99.87	99.62
Artificial Raspberry #59.434/TP 05.50 (Firmenich) or equal	0.10	-
Artificial Raspberry Flavor No. 2397 (Borden) or equal	-	0.35
Color, Red No. 40 (see 6.3)	0.029	0.029
Color, Blue No. 1 (see 6.3)	0.001	0.001

Layer the flavoring and the color between two layers of the fruit soup base and mix until homogeneous.

3.3.4 Pouch filling and sealing. The product shall be filled and sealed in pouches so as to meet the requirements in 3.4 and MIL-P-44276.

3.5 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:10Z
Check the source to verify that this is the current version before use.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. The contents of each pouch shall weigh not less than 50.0 grams and not more than 55.0 grams.

d. Final blended product shall be a fine granular pink product with dark particles from the color ingredients.

e. The moisture content of the product shall not exceed 1.0 percent.

f. Final blended product shall be free from lumps such that product passes easily through a US Standard No. 16 sieve or equivalent.

3.4.1 Palatability. The finished product shall be equal to or better than the preproduction sample (see 6.1) in palatability and overall appearance.

3.5 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110: Current Good Manufacturing Practice in Manufacture, Processing, Packaging or Holding of Human Foods; and the plant sanitation requirements of the appropriate Government inspection agency.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of supervision necessary to assure compliance with the requirements of this document.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:10Z
Check the source to verify that this is the current version before use.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, and filling requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Net weight inspection. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Results shall be reported to the nearest 0.1 gram. Any individual net weight of less than 50.0 grams or more than 55.0 grams shall be scored as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

4.5.4 Product inspection. The finished product shall be examined for the defects listed in table I. The lot size shall be expressed in pouches. The sample unit shall be the contents of one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects.

Source: https://assist.dla.mil -- Downloaded: 202610-22T01:10Z
Check the source to verify that this is the current version before use.

TABLE I. Product defects 1/ 2/

Category	Defect
Major	Dry product
101	Product not of specified color
102	Presence of lumps which do not pass easily through a U.S. Standard No. 16 Sieve or equivalent.

1/ The presence of foreign material (e.g., glass, dirt, insect parts, hair, wood, metal) foreign odor or flavor (e.g., burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance shall be cause for rejection of the lot.

4.5.5 Moisture content testing. The moisture content shall be determined in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists (AOAC); Chapter: Sugar and Sugar Products; Section: Moisture; Method: Drying in Vacuum. The results shall be reported to the nearest 0.1 percent. Any result not conforming to the requirement of 3.5 shall be classified as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-2 and AQL, expressed in terms of defects per hundred units, shall be 1.5.

4.5.6 Packaging inspection. Inspection of packaging shall be in accordance with the quality assurance provisions of MIL-P-44276.

5. PACKAGING

5.1 Packing. Packing shall be in accordance with the requirements of MIL-P-44276.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type of product required (see 1.2).
c. When a first article is required (see 3.1, 4,4 and 6.2).
d. Provisions for approved preproduction samples (see 3.4.1 and 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:10Z
Check the source to verify that this is the current version before use.

6.2 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Color dispersion. To achieve better dispersion of color ingredients during fruit soup dry mix formation, each color may be mixed with dextrose in the ratio of 1 part color up to a maximum of 9 parts dextrose. The percent of fruit soup base in the formula should be decreased accordingly so that the ingredients will total 100 percent.

6.4 Starch. The starch used in developing the product was Gelatinized Dura-Jel Starch manufactured by: A.E. Staley Manufacturing Company, Food Starch Section - Industrial Products, 2200 East Eldorado Street, Decatur, Illinois 62525.

6.5 Subject term (key word) listing.

Cold weather rations
Fruit
Operational ration component
Ration
Soup

Custodians:

Army - GL

Review activity:

Army - MD, TS
Navy - MC, MS, SA
Air Force - 50
DP - SS

Preparing activity:

Army - GL

Project No. 8915-A807

9/10 GL:MR

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:10Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

US ARMY NATICK RESEARCH
and DEVELOPMENT CENTER
ATTN: STRNC-ES
Natick, MA 01760-5014

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Natick Research
and Development Center
ATTN: STRNC-ES
Natick, MA 01760-5014

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:10Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-44282(GL)	Fruit Soup, Artificially Flavored (Operational Ration Component)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

NATICK OP 20 Nov 84

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:10Z
Check the source to verify that this is the current version before use.